

In vivo monitoring laser tissue interaction using high resolution Fourier-domain optical coherence tomography

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ABSTRACT

Laser-induced therapies include laser ablation to remove or cut target tissue by irradiating high-power focused laser beam. These laser treatments are widely used tools for minimally invasive surgery and retinal surgical procedures in clinical settings. In this study, we demonstrate laser tissue interaction images of various sample tissues using high resolution Fourier-domain optical coherence tomography (Fd-OCT). We use a Q-switch diode-pumped Nd:YVO4 nanosecond laser (532nm central wavelength) with a 4W maximum output power at a 20 kHz repetition rate to ablate *in vitro* and *in vivo* samples including chicken breast and mouse ear tissues. The Fd-OCT system acquires time-series B-scan images at the same location during the tissue ablation experiments with 532nm laser irradiation. The real-time series of OCT cross-sectional (B-scan) images compare structural changes of 532nm laser ablation using same and different laser output powers. Laser tissue ablation is demonstrated by the width and the depth of the tissue ablation from the B-scan images.

Keywords: Laser tissue interaction, 532nm laser ablation, ablation efficiency enhancement, in-vivo ablation, OCT real-time monitoring

1. INTRODUCTION

Tissue ablation by medical lasers is widely used in current clinical settings for laser surgery and skin treatment due to precise control of irradiation powers, exposure times as well as other laser parameters. Laser energy is absorbed in target tissue during the treatment and photo-thermal effects by laser-tissue interactions generate removing or cutting tissue structures. Although laser-irradiation therapies are relatively convenient, laser ablation has some risks owing to using high power focused beam. For instance, the high power focused beam could cause collateral tissue damage on the laser exposed region. Several studies of laser tissue ablation have been conducted for minimizing the thermal damage zone and enhancing ablation efficiency for the given laser energy.^{[1], [2]} The pulsed CO₂ laser (10.6μm) tissue ablation study by Walsh et al. demonstrated longer laser pulse exposure created larger thermal damage zones so that short laser pulses resulted in minimal thermal damage in laser ablation. In addition, darker staining areas in skin were damaged more intense by the pulsed CO₂ laser than normal color skin areas. These results were examined by photomicrograph using a conventional light microscope acquired from punch biopsy of laser irradiated sites.

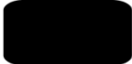
In this research, we use a 532nm (commonly used wavelength for laser treatment) laser to ablate *in vitro* and *in vivo* tissue samples. Moreover, Fourier-domain optical coherence tomography (Fd-OCT) acquires cross-sectional images for monitoring laser tissue interactions during irradiation. We design the 532nm laser and OCT system having a same focal plane using a dichroic mirror and object lens for real-time monitoring of laser ablation. Throughout the experiment, ablation differences by pulse duration are examined as well as laser ablation changes by tissue color types are investigated using qualitative OCT images.

2. METHODS AND MATERIALS

2.1 Tissue samples

Punched tissues of fresh chicken breast and porcine skin are prepared for *in-vitro* sample imaging. Especially porcine skin could substitute human skin in the experiment of percutaneous absorption.^[3] Furthermore, ears of hairless mice and depilated thighs of Mongolian gerbils are used for *in-vivo* monitoring laser tissue interactions. In order to observe laser ablation changes by tissue colors, we spread different color inks on sample tissues using water-based pens (Plus pen 3000, MONAMI, South Korea). As shown in Table 1, four colors (CMYK) are used for this study.

Table 1. Color tables

	color	CMYK	Color code
	Black	C : 0 M : 0 Y : 0 K : 100	#000000
	Blue	C : 100 M : 75 Y : 0 K : 0	#0040FF
	Red	C : 10 M : 100 Y : 100 K : 0	#E60000
	Green	C : 100 M : 0 Y : 100 K : 0	#00FF00

C : Cyan, M : Magenta, Y : Yellow, K : blaK

2.2 Fourier-domain optical coherence tomography (Fd-OCT) and 532nm Nd:YVO4 laser system

In order to investigate laser tissue interaction in real time, Fourier-domain optical coherence tomography (Fd-OCT) is used. OCT is a cross-sectional optical imaging solution used to various applications of medicine including ophthalmology, dermatology and cardiology.^{[4]-[12]} As shown in Figure 1, a superluminescent diode (SLD, SUPERLUM, Carrigtwohill, Ireland, central wavelength = 860, bandwidth = 135nm, output power = 3mW) with an isolator is used as a broadband light source. A diffraction grating (1200L/mm at 840nm, Wasatch Photonics, Inc. Durham, NC, USA), a focusing lens and a line scan camera (spL4096-140km, Basler AG, Ahrensburg, Germany) are used to acquire spectral data. A reference arm is comprised of a neutral density filter wheel, 30mm focal lens and crystals for compensating dispersion (Thorlabs, INC.). A sample arm including dichroic mirror (Semrock, INC, Rochester, New York, USA) and 2-Axis galvo scanner (Cambridge Technology, Bedford, MA USA) is designed to match a same focal plane with the ablation laser source. Each part of OCT is connected by the fiber coupler (OZ, Optics, Ontario, Canada) with 50/50 splitting ratio as the Michelson interferometer configuration.

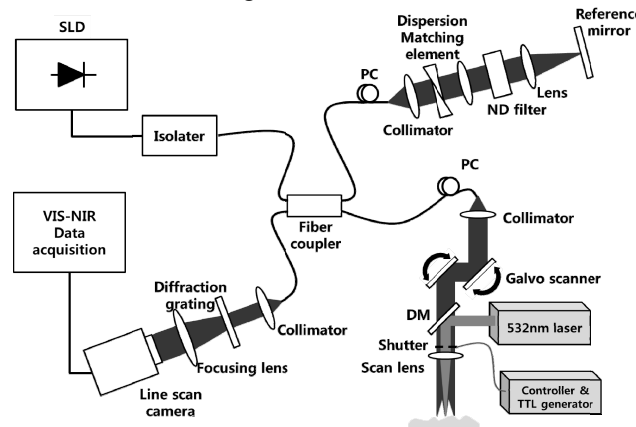


Figure 1. Fd-OCT & laser system for real-time ablation monitoring. SLD : superluminescent diode, Dispersion Matching element : same property achromatic lens with scan lens and crystal pair, ND filter : neutral density filter, PC : polarization controller, DM : dichroic mirror

Table 2. Difference (loss rate) between laser output and lens output Unit : Watts

Laser output	Lens output	Laser to Lens loss rate
0.122 W	0.069 W	43.44%
0.286 W	0.165 W	42.31%
0.515 W	0.295 W	42.72%
0.807 W	0.46 W	43.00%
1.192 W	0.695 W	41.69%
Average of laser to lens loss rate		<u>42.9%</u>

The Q-switch diode-pumped Nd:YVO₄ nanosecond laser is used to ablate sample tissues. This laser is operated in 15ns pulse duration, a 20kHz repetition rate, 532nm wavelength, and 1.6mm Gaussian beam diameter aperture. The laser output power and target plane power are measured by a laser power meter (FieldMax II-TO, Coherent, Santa Clara, CA, USA) shown in Table 2. The precise output wavelength of the laser is measured as shown in Figure 2.

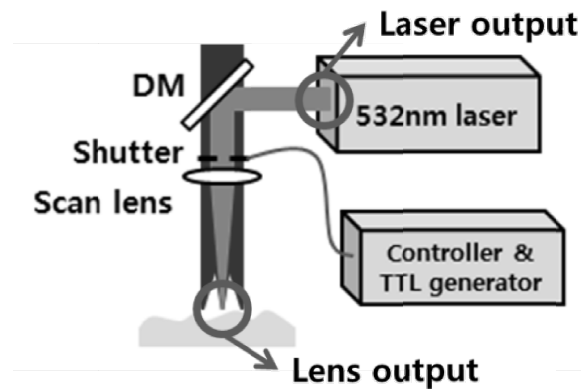


Figure 2. Position of measuring laser powers. The lens output power is measured at a focal length of the scan lens (30mm)

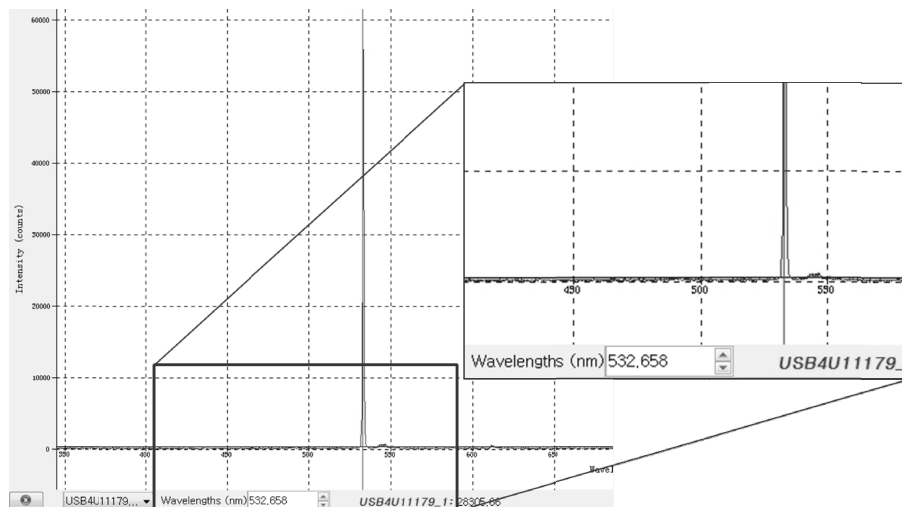


Figure 3. Measured spectrum of the ablation laser by the spectrometer (USB4000, ocean optics, Dunedin, FL, USA)

To modulate accurate exposure time and pulse duration, the automatic shutter (Thorlabs, INC. Newton, New Jersey, USA) is used consisting of an optical beam shutter as well as a shutter controller modulated by a custom external TTL signal generator using a microcontroller unit (Arduino UNO).

3. RESULTS

3.1 Pulse modulated ablation

Pulse duration is modulated by a custom TTL generator and the shutter through S/W coding. As shown in Figure 4, we could observe ablation differences between (b) and (d) in same energy exposure ($481\text{mW} * 8\text{s} = 3.848\text{J}$) with different laser pulse durations (e) and (f), respectively. Especially, (b) shows a surface of sample is not destroyed.

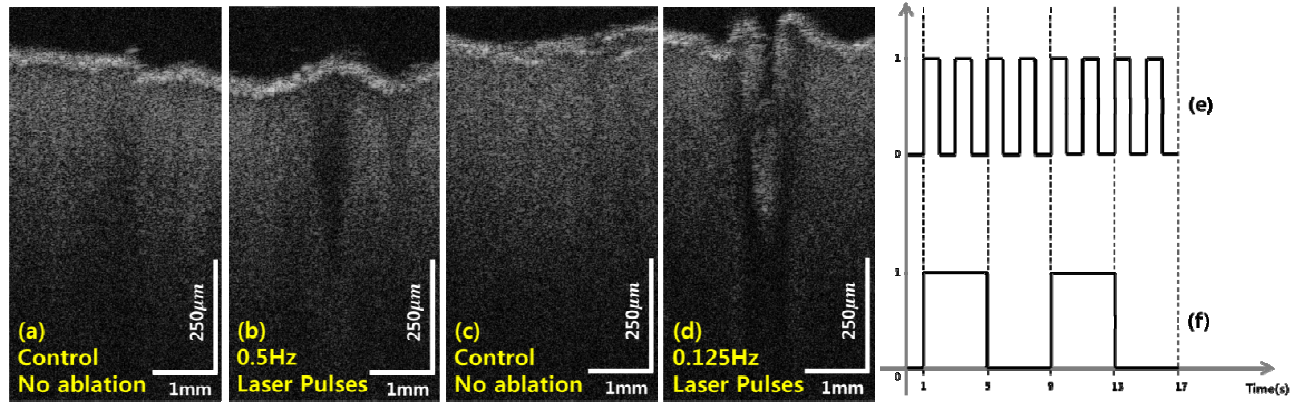


Figure 4. Ablation differences by pulse durations. (a) : a control image of (b), (b) : 0.5Hz laser irradiation (1sec irradiation and 1sec break) for total 16sec, (c) : a control image of (d), (d) : 0.125Hz laser irradiation (4sec irradiation and 4sec break) for total 16sec. (e) : 0.5Hz laser pulses used in (b), (f) : 0.125Hz laser pulses used in (d). 1 means shutter open and 0 means shutter closed at the Y-axis in the graph.

3.2 Laser ablation changes by tissue colors

The chicken breast and porcine skin are used for taking in-vitro images. The chicken breast can be substituted by the human muscle and the porcine skin also can be substituted by the human skin.

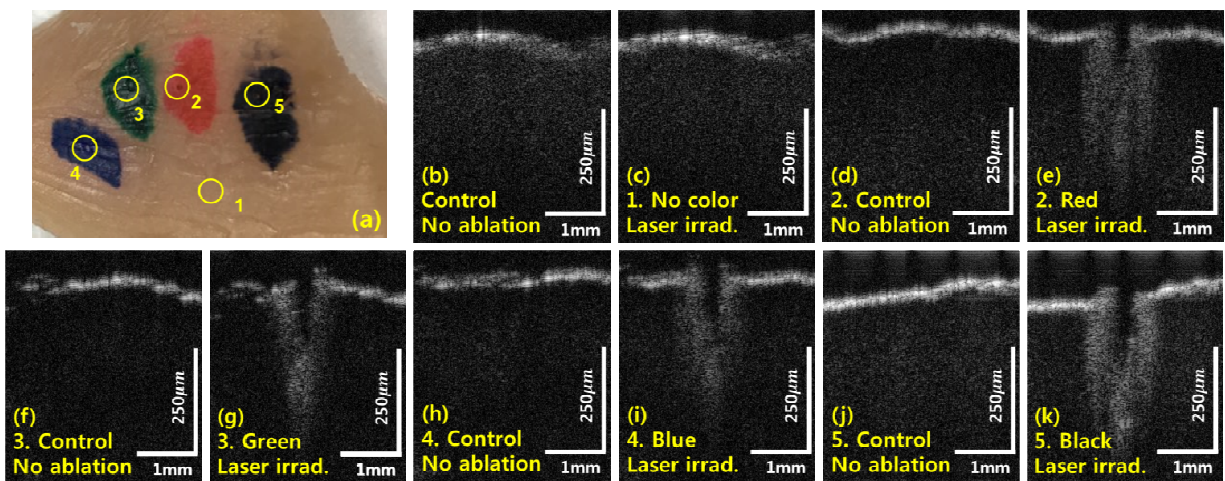


Figure 5. Laser ablation on the chicken breast. OCT images of (b), (d), (f), (h), (j) are from control samples and (c), (e), (g), (i), (k) are from laser irradiated samples. 532nm laser ablation (799mW) for 1 sec. (b), (c) : Non-colored samples. (d), (e) : red colored samples. (f), (g) : green colored samples. (h), (i) : Blue colored samples. (j), (k) : Black colored samples.

Laser beam of 685mW power is irradiated for 3sec on the porcine skin thicker than the human skin.

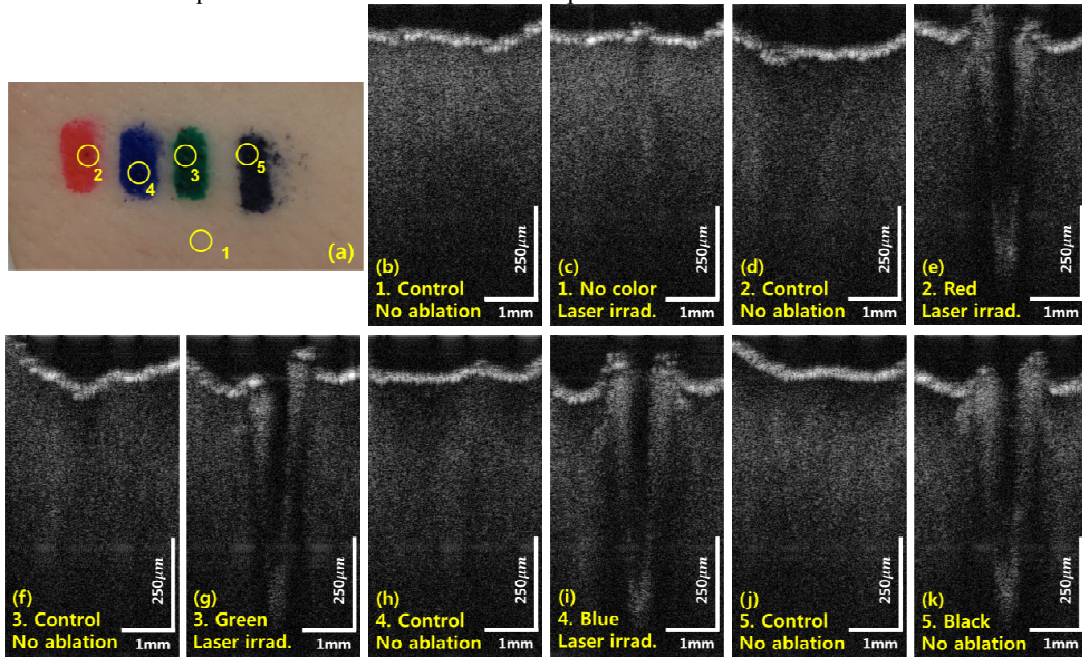


Figure 6. Laser ablation on the porcine skin. OCT images of (b), (d), (f), (h), (j) are from control samples and (c), (e), (g), (i), (k) are from laser irradiated samples. 532nm laser ablation (685mW) for 3 sec. (b), (c) : Non-colored samples. (d), (e) : Red colored samples. (f), (g) : Green colored samples. (h), (i) : Blue colored samples. (j), (k) : Black colored samples.

As shown in Figure 7, images after irradiation show volcano shapes of thermal contraction on the thigh skin of a Mongolian gerbil.

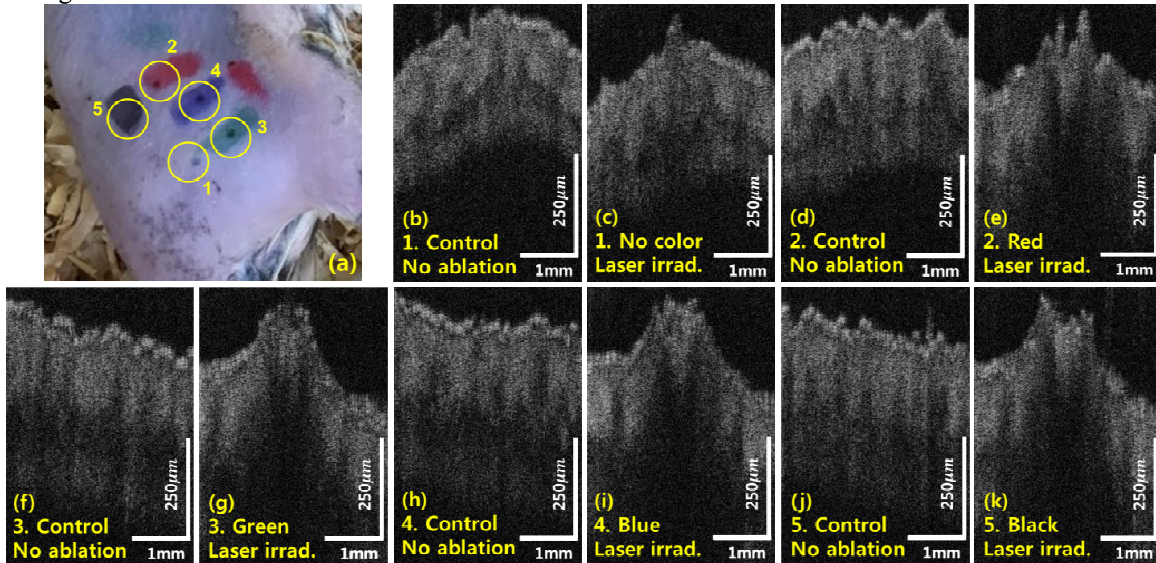


Figure 7. Laser ablation on the Mongolian gerbil. OCT images of (b), (d), (f), (h), (j) are from control samples and (c), (e), (g), (i), (k) are from laser irradiated samples. 532nm laser ablation (685mW) for 1 sec. (b), (c) : Non-colored samples. (d), (e) : Red colored samples. (f), (g) : Green colored samples. (h), (i) : Blue colored samples. (j), (k) : Black colored samples.

4. DISCUSSION AND CONCLUSIONS

High resolution Fd-OCT provides cross-sectional images of laser tissue interactions with the Nd:YVO4 nanosecond 532nm laser. In order to observe ablation differences by pulse duration, we compare OCT B-scan images from control tissue and thermally-damaged-tissue by the pulse modulated laser (repeat 1 sec irradiation and 1 sec break as well as 4 sec irradiation and 4 sec break for total 16 sec). These experiments demonstrate long pulse irradiation generates more thermal damages and deeper craters than short pulse irradiation. In addition to investigating ablation differences by skin colors, ablation images of four colored samples (red, green, blue, and black) are used. Although the laser irradiation on the non-colored samples is not enough to ablate non-colored tissues, the colored samples cause thermal damages and craters on the ablated areas.

5. ACKNOWLEDGEMENTS

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